Best-Fit

Input:

user07@jannani:~ — /usr/bin/vim bestfit.c

```
#include<stdio.h>
int main(){
        int m,n;
        printf("Enter number of memory blocks:\n");
        scanf("%d",&m);
        int blockSize[m],remaining[m];
        printf("Enter size of each block:\n");
        for(int i=0;i<m;i++){
                scanf("%d",&blockSize[i]);
                remaining [i]=blockSize[i];
        }
        printf("Enter number of processes:\n");
        scanf("%d",&n);
        int processSize[n];
        printf("Enter size of each process:\n");
        for(int i=0;i<n;i++){
                scanf("%d",&processSize[i]);
        }
        for(int i=0;i<n;i++){
                int bestIndex=-1;
                for(int j=0;j<m;j++){
                        if(remaining[j]>=processSize[i]){
                                if(bestIndex == -1  || remaining[j]<remaining[bestIndex]){
                                        bestIndex=j;
                                }
                                if(bestIndex != -1){
                                        int frag=remaining[bestIndex]-processSize[i];
                                        printf("Process %d of size %d is allocated to Block %d of size %d with fragement %d\
n", i+1,processSize[i],bestIndex+1,blockSize[bestIndex],frag);
                                        remaining[bestIndex]-=processSize[i];
                                }
                                                                                                                1,17          Top
                                }
                                else{
                                        printf("Process %d of size %d is not allocated\n",i+1,processSize[i]);
                                }
                        }
                }
        }
        return 0;
-- INSERT --                                                                                            38,2-9        Bot
```

Output:

```
user07@jannani:~
[user07@jannani ~]$ gcc bestfit.c
[user07@jannani ~]$ ./a.out
Enter number of memory blocks:
2
Enter size of each block:
150 300
Enter number of processes:
3
Enter size of each process:
30 70 150
Process 1 of size 30 is allocated to Block 1 of size 150 with fragement 120
Process 1 of size 30 is allocated to Block 1 of size 150 with fragement 90
Process 2 of size 70 is allocated to Block 1 of size 150 with fragement 20
Process 2 of size 70 is allocated to Block 1 of size 150 with fragement -50
Process 3 of size 150 is allocated to Block 2 of size 300 with fragement 150
[user07@jannani ~]$
```